

Progressive Mathematics Initiative Mathematics Curriculum Chapter Plan	
Title: Ratios, Proportions & Percents	
Grade Level: 6	Length of Time: 6 weeks
Chapter Summary: This chapter will introduce formally the concepts of ratios, proportions, and percent problems. They will review definitions about ratios, develop a sense of converting between different measurements, and work with unit rate problems. They will then be able to solve problems involving percents and use that knowledge in real-world situations involving them.	
Learning Targets	
PARCC ■ Major Clusters; ■ Supporting Clusters; ● Additional Clusters	
Domain: Ratios and Proportional Relationships	
Cluster: Understand ratio concepts and use ratio reasoning to solve problems.	
Standard #:	Standard:
6.RP.1	Understand the concept of a ratio and use ratio language to describe a ratio relationship between two quantities. For example, “The ratio of wings to beaks in the bird house at the zoo was 2:1, because for every 2 wings there was 1 beak.” “For every vote candidate A received, candidate C received nearly three votes.”
6.RP.2	Understand the concept of a unit rate a/b associated with a ratio $a:b$ with $b \neq 0$, and use rate language in the context of a ratio relationship. For example, “This recipe has a ratio of 3 cups of flour to 4 cups of sugar, so there is $3/4$ cup of flour for each cup of sugar.” “We paid \$75 for 15 hamburgers, which is a rate of \$5 per hamburger.”
6.RP.3	Use ratio and rate reasoning to solve real-world and mathematical problems, e.g., by reasoning about tables of equivalent ratios, tape diagrams, double number line diagrams, or equations. a. Make tables of equivalent ratios relating quantities with whole number measurements, find missing values in the tables, and plot the pairs of values on the coordinate plane. Use tables to compare ratios. b. Solve unit rate problems including those involving unit pricing and constant speed. For example, if it took 7 hours to mow 4 lawns, then at that rate, how many lawns could be mowed in 35 hours? At what rate were lawns being mowed? c. Find a percent of a quantity as a rate per 100 (e.g., 30% of a quantity means $30/100$ times the quantity); solve problems involving finding the whole, given a part and the percent. d. Use ratio reasoning to convert measurement units; manipulate and transform units appropriately when multiplying or dividing quantities
Domain: Standards for Math Practice	
Standard #:	Standard:
MP1	Making sense of problems and persevere in solving them.
MP2	Reason abstractly and quantitatively.
MP3	Construct viable arguments and critique the reasoning of others.
MP4	Model with mathematics.
MP5	Use appropriate tools strategically.
MP6	Attend to precision.
MP7	Look for and make use of structure.
MP8	Look for and express regularity in repeated reasoning.
Chapter Essential Question: • Is it important to know how to solve for unit rates? • What is the connection between a ratio and a	Chapter Enduring Understanding: • Reasoning about ratios and proportions will help solve real-world situations. • The relationships between fractions, decimals,

fraction/decimal? • How are ratios used in the real world? • Where can examples of ratios and rates be found? • What does a percent represent? • How can knowledge about percents aid me in real-world situations?	and percents are critical and needed to solve problems.
Chapter Objectives: • <i>Students will be able to use ratios to describe proportional situations.</i> • <i>Students will be able to represent ratios and percents with concrete models, fractions, and decimals.</i> • <i>Students will be able to apply their knowledge of ratios and proportions to percent problems.</i> • <i>Students will be able to solve problems involving percents.</i> • <i>Students will be able to make conversions between different measurements and unit ratios.</i>	
Evidence of Learning	
Formative Assessments: • SMART Response questions used throughout the chapter. • 5 Quizzes	
Summative Assessment: • Chapter Test • Chapter Project	
Lesson Plan	
Topics	Timeframe
Topic #1: Writing Ratios Lab: RAFT – Salmon You Can Count On	2 days
Topic #2: Equivalent Ratios Lab: PhET Proportion Playground Quiz #1	3 days
Topic #3: Rates & Unit Rates <i>Select one of the labs below:</i> Lab: RAFT – Happy Trails Mix Lab: PhET Unit Rate	3 days
Lab: Design on a Dime Project	2 days
Topic #4: Using Ratios to Convert Measurements Quiz #2	3 days
Topic #5: Converting Unit Ratios Quiz #3	3 days
Topic #6: Percents & Fractions	3 days
Topic #7: Percents & Decimals Quiz #4	2 days
Topic #8: Using Percents Quiz #5	4 days
Lab: Orange Soda Experiment	3 days
Review and Chapter Test	2 days
Curriculum Development Resources: • https://njctl.org/courses/math/6th-grade-math/ratios-proportions-percents/ • http://www.raftbayarea.org/ideas/Salmon%20You%20Can%20Count%20On.pdf • https://phet.colorado.edu/en/simulation/proportion-playground • http://www.raftbayarea.org/ideas/Happy%20Trails%20Mix.pdf • https://phet.colorado.edu/en/simulation/unit-rates	

Lesson Components	
21st Century Skills	
• Financial, Economic, Business, and Entrepreneurial Literacy	
21st Century Themes	
• Critical Thinking and Problem Solving • Communication and Collaboration • Life and Career Skills	